26021 RTOS5 - Mastering USB communication: Develop USB Device applications with Zephyr's USB Device Stack on your custom board

Introduction

USB is now a standard serial communication channel to connect embedded systems to PCs. Upon completion of the course, you will be able to create an enumerable USB device and transfer data between a Microchip MCU and a PC using Zephyr's USB Device Stack. The hands-on part of this class is based on a development board that is not yet supported by Zephyr tree, and you will also learn how to add your own board to a Zephyr based application.

Upon completion, you will be able to:

- Create your custom board support and compile the Zephyr blinky example on it.
- Add USB CDC ACM class to the example and communicate with the PC using a terminal.
- Transfer generic data between MCU and PC applications.

This class includes four labs:

Lab 1 - Run the Blinky sample project on your custom board

Lab 2 - Redirect the Console to the USB

Lab 3 - Build a CDC device and communicate with a terminal on PC

Lab 4 - Implement a simple command-response protocol

Prerequisites

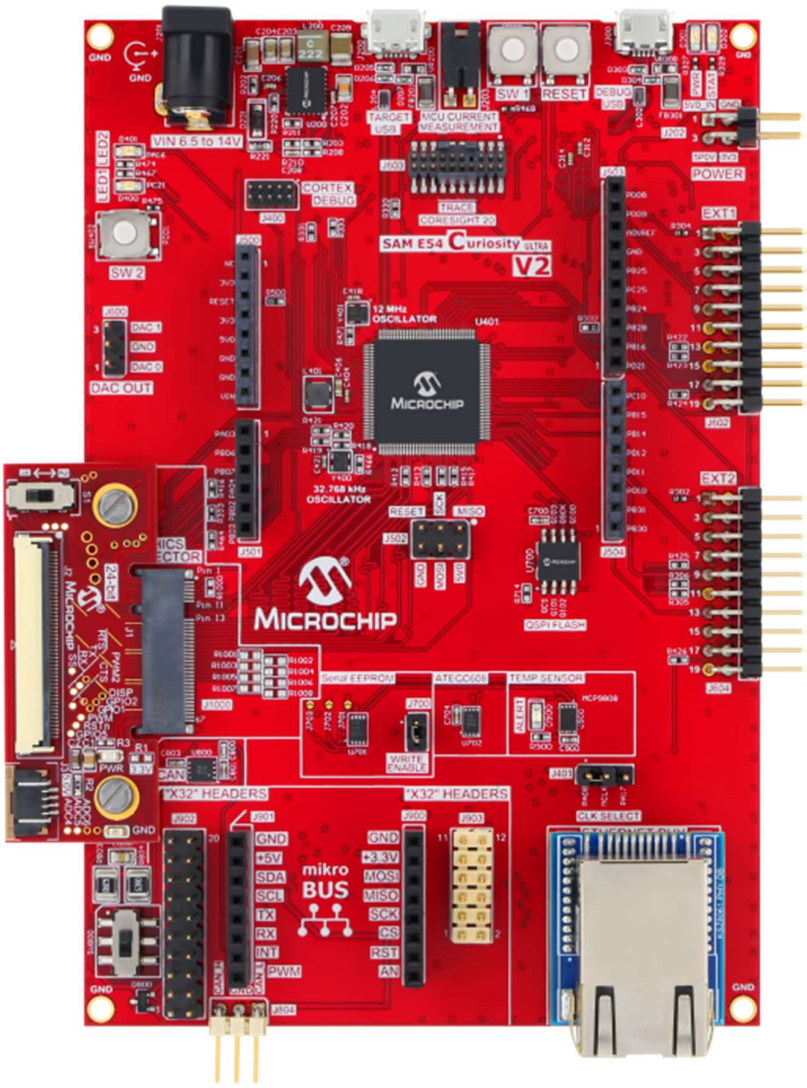
The lab material assumes you have prior experience with:

- Zephyr RTOS environment
- C language programming

Hardware Requirements

Here is the list of hardware needed to complete the labs:

- SAM E54 Curiosity Ultra Development Board v2: EV66Z56A



- 2x CABLE, USB A TO MICRO-B

Software Requirements

Here is the list of software needed to complete the labs:

- Microsoft Visual Studio Code® (v1.110 or Later) with following extensions:
  - C/C++ Extension Pack
  - Serial Monitor
  - Cortex-Debug
  - Python
  - Python Environments

Zephyr OS will be installed in C:\MASTERs\26021_RTOS5 folder running the installation script you can download from:
https://github.com/gcolombo-mchp/26021_RTOS5/blob/main/Scripts/install_26021_RTOS5.ps1